HotSpotter: predicting energetic hot spots at protein–protein interfaces from a single structure

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Abstract

Background. The binding energy of a protein–protein interface is not distributed evenly across its contact surface. A minority of residues — hot spots — account for most of it, and identifying them is the starting point for designing an inhibitor of an interaction or engineering a tighter binder. The standard structural shortcut is burial: rank residues by how much surface area they bury on complex formation. Burial measures contact area, not energy.

Methods. HotSpotter ranks interface residues from a single crystal structure using gradient-boosted trees over 30 hand-built features spanning interaction chemistry, burial and topology, van der Waals packing, evolutionary conservation, and side-chain mobility. It was trained on 1,570 alanine-scanned residues across 172 complexes from SKEMPI 2.0, with a hot spot defined as $\Delta\Delta G \geq 2.0\,\mathrm{kcal/mol}$. Cross-validation is grouped by complex; a sequence-identity-clustered variant and nested hyperparameter selection are reported alongside.

Results. HotSpotter reaches PR-AUC 0.5629 against naive burial's 0.3927 (+43% relative) and ROC-AUC 0.7936 against 0.6639. Per complex, its top-ranked residue is a measured hot spot 78.2% of the time against burial's 60.0%. On 349 externally scored mutations it beats BeAtMuSiC with a bootstrap confidence interval excluding zero (+0.172 PR-AUC) and is statistically indistinguishable from Robetta and two published SVMs. Performance is essentially unchanged on structures perturbed by 1 Å of rigid-body docking error.

Conclusions. A carefully evaluated tabular model over interpretable structural features performs comparably to published hot-spot predictors that use more expensive physics, without molecular dynamics or free-energy calculation. The dominant remaining error is not model capacity but information a single bound structure does not contain.

1 Background

Protein–protein interactions carry out most cellular processes, and interactions that go wrong underlie a great many diseases. Interfaces are large and comparatively flat, which is why they were long considered poor drug targets. The observation that reframed them is that their binding energy is concentrated: alanine-scanning mutagenesis of the human growth hormone receptor interface showed that a small subset of residues accounted for most of the binding free energy, while the majority contributed little [1]. Those residues — hot spots — are a small-molecule-sized target inside an otherwise undruggable surface, and they are what protein engineering must strengthen to tighten a binder.

Identifying hot spots experimentally means alanine-scanning mutagenesis: substituting one residue at a time and measuring the change in binding free energy ($\Delta\Delta G$). The method is definitive and slow. A typical interface has 40–50 residues; scanning all of them is rarely done. In the dataset used here, the median interface has 45 residues and SKEMPI records measurements for a median of 31% of them. Computational prediction is therefore not a replacement for the experiment but a way to order it — which residues to test first.

The simplest structural predictor is burial, usually change in solvent-accessible surface area on complex formation (ΔSASA), as reported by tools such as PISA. It is cheap, requires only the complex structure, and is correlated with $\Delta\Delta G$ because residues that bury more surface tend to

make more contacts. It is also the wrong quantity. Area is not energy. A large side chain can bury substantial surface in loose, non-specific packing and cost almost nothing to delete, while a smaller residue held by a buried salt bridge or a hydrogen-bond network is expensive to lose. Burial also cannot see conservation, side-chain rigidity, or the occlusion from solvent that the O-ring hypothesis identifies as characteristic of hot spots [2].

This project asks a narrow, measurable question: *starting from one bound structure, how much better than burial can a model do using only features that are cheap to compute, and how confident can we be in that number?* The second half of the question received as much work as the first.

2 Methods

2.1 Data and labels

Labels come from SKEMPI 2.0 [3], a curated database of experimentally measured changes in binding free energy on mutation. Only **single point mutations** were used; multiple mutations were excluded throughout, because their $\Delta\Delta G$ cannot be attributed to one residue.

Mutations were collapsed to one row per residue position. The primary label strategy keeps only X→Ala substitutions, which matches the literature definition of a hot spot and makes the label mean one specific thing, at the cost of discarding data. Where a position carried several measurements, the largest $\Delta\Delta G$ was taken. The collapse key is (PDB entry, partner chains, chain, residue number, insertion code); omitting the insertion code merges distinct residues, since Kabat-numbered chains place several at one number — 1EAW chain A has four residues numbered 60.

A residue is labelled a hot spot when $\Delta\Delta G \geq$ **2.0 kcal/mol**, the Clackson–Wells / Bogan–Thorn convention. The resulting training set is **1,570 residues across 172 complexes, 362 of them hot (23.1%)**.

Structures were downloaded from the RCSB PDB. Interfaces were detected by heavy-atom contact within 5.0 Å using a KD-tree.

2.2 Features

Thirty features per residue, in six groups:

Group	Cols	What it measures
Burial / accessibility	6	SASA bound and unbound, ΔSASA, relative SASA, interface membership
Interface topology	5	cross-interface contacts, atom contacts, neighbours, packing density, centrality
Residue identity	7	charge, hydropathy, volume, flexibility, aromatic / charged / polar flags
Interaction chemistry	6	salt bridges, hydrogen bonds, hydrophobic and aromatic contacts, totals
Van der Waals packing	3	smoothed Lennard-Jones side-chain energy, pair count, repulsive term
Conservation / mobility	3	ConSurf-DB conservation score, B-factor, side-chain / backbone B-factor ratio

Two choices in this set are worth stating explicitly.

Chemistry counts are distinct partner residues, not atom pairs. Counting atom pairs rewards large residues twice, once for size and once for contact.

Side-chain mobility is expressed as a ratio, not a raw value. The mean B-factor over a whole residue is dominated by its backbone, which mostly reports secondary structure — a different phenomenon from side-chain plasticity, which is what should matter for a deletion. The raw side-chain B-factor does not isolate it either, because the absolute B scale varies between structures with resolution and refinement protocol. Dividing by the residue's own backbone B-factor cancels that per-structure scale. Measured, the ratio is worth +0.0161 **PR-AUC** ($p = 0.0009$) and the raw side-chain value nothing at all; the two are nearly uncorrelated ($r = -0.06$), so the ratio is new information rather than a restatement.

Missing values are passed through rather than imputed. ConSurf-DB has no entry for about 4% of chains, and those chains are not a random sample — they are short peptides, 50.9% hot against a 23.4% base rate. The missingness is informative. Gradient-boosted trees learn a default branch direction per split, which handles this natively; median-filling the column instead costs 78% of the feature's measured benefit. Model classes that cannot accept missing values are given an imputed column plus a binary missing-indicator, with imputation values fitted on the training fold only.

Two columns are computed and stored but deliberately **excluded** from the model: crystallographic resolution and experimental method. Both measured as significantly harmful (resolution -0.0137, $p = 0.011$; both together -0.0160, $p = 0.003$). The reason is structural rather than statistical: each takes exactly one value per complex, so it cannot reorder residues within an interface, which is the only thing that helps a user. It can only shift whole complexes against each other in the pooled metric, and that does damage.

2.3 Model

XGBoost [4] with 300 trees, maximum depth 4, learning rate 0.05, row and column subsampling at 0.8, and L1 regularization `reg_alpha` = 50. The positive class is weighted by the negative-to-positive ratio of the training fold. Predictions are averaged over three random seeds, because the subsampling makes a single seed noisy; six seeds were tested and bought +0.0016, which is not worth the fits.

The configuration is frozen in one place and imported by every script that fits a model. This is not tidiness. Unlike scikit-learn's `class_weight="balanced"`, XGBoost's `scale_pos_weight` depends on the training labels and so cannot live in a static configuration dictionary; a comparison harness that forgets it silently runs the shipped model unbalanced against balanced rivals. That happened twice during development and both result tables had to be discarded.

2.4 Evaluation

Folds are grouped by complex. No residue is scored by a model that has seen another residue from the same interface. Interface residues share burial, partner context and structure, so a random residue split leaks badly; it is the easiest way to fool yourself in this problem and is a plausible source of the spread in published numbers. A fixed fold assignment is stored in the repository so every experiment is comparable.

The primary metric is PR-AUC, because the positive rate is 23% and accuracy is useless at that imbalance. ROC-AUC is reported alongside. Per-complex **hit rate at** k — would a biologist choosing k residues to mutate hit a real hot spot? — is reported separately from **precision at** k, because conflating them overstates the result.

Bootstraps resample complexes, not residues. Residues within one interface are correlated,

so resampling residues understates the interval.

Hyperparameter selection bias is measured, not assumed. `reg_alpha` = 50 was originally chosen by sweeping eight values over the same folds the score is reported on, which is selection bias. Nested cross-validation — an inner GroupKFold over each outer fold's training complexes only — puts the honest estimate at **0.5550 against the reported 0.5629**, an optimism of +0.0079 **(paired $p = 0.036$)**. That is comparable to a whole feature's contribution, so it is disclosed rather than absorbed, and the value has been frozen since.

3 Results

3.1 Against naive burial

Five folds grouped by complex, five seeds.

	Naive burial (ΔSASA)	HotSpotter
PR-AUC	0.3927	**0.5629**
ROC-AUC	0.6639	**0.7936**
ROC-AUC, literature label definition	0.7020	**0.8625**
PR-AUC, literature label definition	0.6568	**0.8174**
Hit rate, top 1	60.0%	**78.2%**
Hit rate, top 3	80.0%	**93.3%**
Hit rate, top 5	84.0%	**96.0%**
Precision at 3	47.8%	**59.6%**

The lift over burial is +0.1702 **PR-AUC, 43% relative**. The literature definition drops the grey zone (hot > 2.0, non-hot < 0.4), which is how published AUCs are computed; reporting an all-rows AUC against a filtered one would flatter this work.

3.2 What each component contributes

Addition	ΔPR-AUC	Significance
L1 regularization	+0.0212	largest single model-side gain
Side-chain / backbone B-factor ratio	+0.0161	$p = 0.0009$, positive in 6 of 6 fold partitions
Van der Waals packing	+0.0115	significant across six fold partitions
ConSurf conservation	+0.0085	—

Van der Waals packing is the only one of the four classical force-field terms that added anything the contact counts were not already capturing.

3.3 What the fold grouping is worth

Holding data, features, model and seeds fixed and varying only how folds are grouped:

Grouping	PR-AUC	Estimates performance on
Random residue split	0.5675	an interface already partly scanned
By complex	0.5560	a new complex
By ≥70% chain-identity cluster	0.5585	a new complex with no close relative in training

Keeping homologous complexes out of the same fold costs nothing measurable. The differences sit inside fold-to-fold noise, so the model is not relying on near-duplicate entries — a real concern, since SKEMPI is redundant and the same pair is often deposited several times.

3.4 Model class comparison

Three alternatives were tested at matched features and matched folds.

Graph neural network. An edge-aware GATv2 [12] over the interface contact graph, with a Wide & Deep skip connection, reached PR-AUC 0.5076 (mean of five seeds) on the 26 node features. The matched XGBoost on those same 26 features and folds scores **0.5249**. With a fold-to-fold spread of ±0.075 this is a tie, with the tabular model nominally ahead. The simpler model was kept. The graph model was never given the three features added later (packing, conservation, mobility), so whether a GNN would close the full gap is untested; the two feature additions it did receive — an ESM-2 conservation scalar and ESM-2 embeddings — both made it worse (0.4978 and 0.4940).

XGBoost + extra-trees ensemble. Blending the two scored +0.0162 PR-AUC over XGBoost alone ($p = 0.003$) when folds are grouped by complex, and the gain survived a model-count-matched baseline of six XGBoost seeds (+0.0146, $p = 0.007$), so it was not simply the effect of averaging more models. It did not survive an honest split. Grouping folds by ≥70% chain identity reverses the sign: the ensemble **loses** by −0.0174 ($p = 0.003$), and extra trees alone falls from 0.5608 to 0.5120 while XGBoost barely moves. Its entire contribution depended on near-duplicate complexes landing in different folds. Two further findings sealed it: on macro PR-AUC — ranking within one interface, averaged over complexes, which is the only ranking a user ever sees — the two are indistinguishable (+0.0034 and +0.0027), so the ensemble's advantage lives in between-complex calibration nobody reads; and extra trees is three times more fragile when conservation is unavailable (−0.0291 against −0.0080), which is exactly the situation a docked or novel complex presents. The ensemble was adopted and then reverted.

Linear models. Best preprocessed result 0.5499 against XGBoost's 0.5561 on identical folds. Scaling choice barely mattered — standard, quantile and power transforms within 0.003.

3.5 External benchmarks

All methods scored on **the same 349 measured alanine mutations across 20 complexes** from Lise et al. [5], with identical labels ($\Delta\Delta G \geq 2.0$ is hot, which is both their threshold and ours), each published method held at its authors' own operating point. HotSpotter is strictly out-of-fold, with the test complex and everything ≥70% identical to it removed from training.

Method	Rows	PR-AUC	ROC-AUC	MCC
mCSM-PPI2 [8] (see below)	344	0.916	0.963	0.74
HotSpotter	349	**0.660**	**0.851**	0.47
Lise SVM [5]	349	0.566	0.788	0.44
Lise TSVM [5]	349	0.565	0.789	0.48
Robetta [6] ($\Delta\Delta G \geq 1.8$)	349	0.531	0.757	0.38
BeAtMuSiC [7]	278	0.454	0.716	0.28
Naive buried area	349	0.438	0.691	0.28

Paired bootstrap over complexes gives +0.172 **PR-AUC** [+0.075, +0.285] **against BeAtMuSiC**, the only comparison whose interval excludes zero. Against Robetta (+0.062), the Lise SVM (+0.029) and the Lise TSVM (+0.031), every interval spans zero. **Twenty complexes cannot**

separate these methods, and the honest statement is that HotSpotter is competitive with them, not better.

Two observations are worth more than the ranking.

BeAtMuSiC scores 0.454 against naive burial's 0.438. A published $\Delta\Delta G$ predictor is barely separable from ranking residues by how much surface they bury. Its interface filter is itself a burial threshold — median buried area 3.3 Å^2 among the residues it declined to score, against 42.1 Å^2 among those it scored — and on barnase–barstar it therefore dropped Arg87, which buries 3.3 Å^2 and has a measured $\Delta\Delta G$ of 5.5 kcal/mol. This project's thesis arrives here as a measurement about somebody else's tool.

The mCSM-PPI2 number cannot be interpreted. Its published training set, S4169, contains **1,559 of our 1,570 rows (99.3%)**, and not one of the 172 complexes falls outside it. A model evaluated on 99.3% of its own training data is not being benchmarked. Their blind CAPRI sets are the only clean route and were not pursued. The 0.916 is reported for completeness and should not be compared with anything.

3.6 Robustness to coordinate error

Because the intended use is a structure the user supplies, which may be a docked model, two error types were simulated separately: rigid-body rotation and translation of one partner, which is what docking error looks like, and independent per-atom Gaussian jitter, which is coordinate uncertainty from low resolution or relaxation.

Input	PR-AUC	Hit@1	Hit@3
Crystal	0.842	94.1%	100%
Rigid 1.0 Å	0.818	94.1%	100%
Rigid 3.0 Å	0.781	79.4%	100%
Jitter 1.0 Å	0.788	85.3%	97.1%
Jitter 3.0 Å	0.681	58.8%	88.2%

At 1 Å interface RMSD — a good docking model — performance is essentially unchanged. The interesting result is that **docking error hurts less than coordinate noise**: rigid 3 Å costs 0.061 PR-AUC, jitter 3 Å costs 0.161, nearly three times more. Rigid error changes which residues sit at the interface while leaving the local geometry of the survivors intact; jitter corrupts the packing, burial and van der Waals terms the features actually measure. The tool needs a well-refined structure more than a perfectly posed one. One caveat: the identity of the top-5 set churns faster than accuracy falls (52% retained at 3 Å), so accuracy is the claim to quote, not list stability.

3.7 Case studies

Two complexes, each removed from training along with its whole ≥70% identity cluster before scoring. Ranks are over the residues SKEMPI measured, because an unmeasured residue in a method's top k is neither a hit nor a miss.

3M62 (Ufd2–Rad23). 33 interface residues, 17 measured, 4 hot (Figure 1). Burial finds **none** of the four in its top four; HotSpotter finds **three**. Tyr97 is the strongest hot spot in the interface ($\Delta\Delta G$ 4.48 kcal/mol) and only the 14th most buried of the 17 measured; HotSpotter ranks it first. The most buried of those 17, Phe9, is not a hot spot at all ($\Delta\Delta G$ 0.99).

1EMV (colicin E9 DNase–Im9). The most completely scanned interface available, so little is unknown: 32 of 43 interface residues measured, 9 hot (Figure 2). Burial finds 4 of 9 in its top

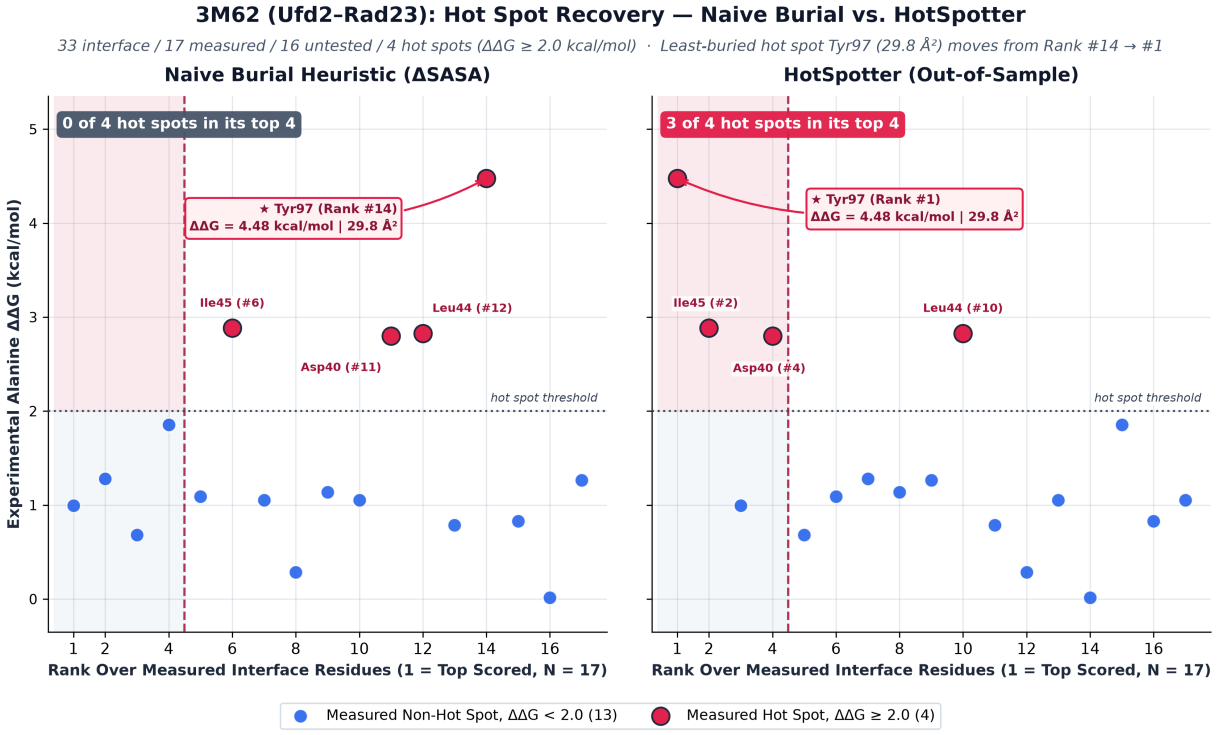


Figure 1: **Hot-spot recovery on 3M62 (Ufd2–Rad23).** Each panel ranks the 17 measured interface residues left to right by one method's score; the y-axis is the experimentally measured $\Delta\Delta G$, so hot spots (red, $\Delta\Delta G \geq 2.0$ kcal/mol) should appear as far left as possible. The shaded band marks each method's top four, the number of hot spots present. Burial places none of the four there; HotSpotter places three. Tyr97, the strongest hot spot in the interface, buries the least surface of the four and is the clearest case of the two quantities disagreeing.

nine; HotSpotter finds 6. Asn75 is the least buried hot spot (25.4 Å^2, $\Delta\Delta G$ 2.33) and moves from 15th by burial to 3rd.

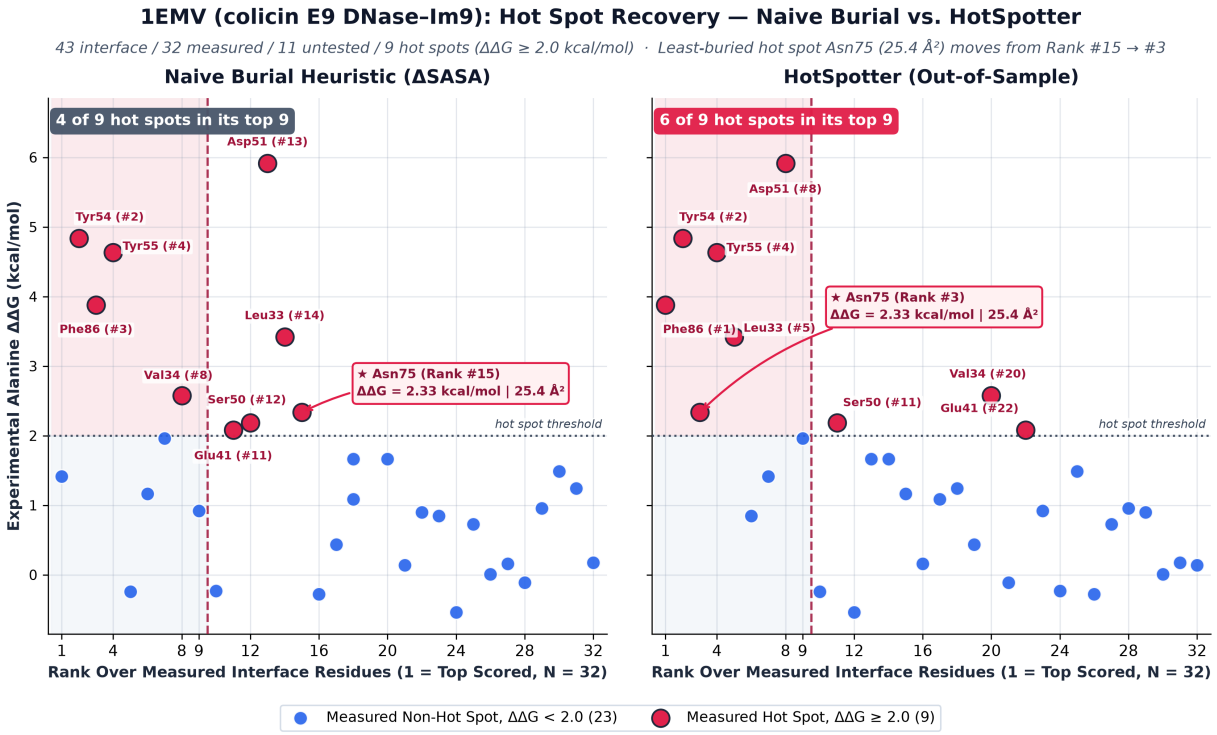


Figure 2: **Hot-spot recovery on 1EMV (colicin E9 DNase–Im9).** Axes as in Figure 1, with the band marking each method's top nine. This interface is the better test of the two because 32 of its 43 residues were measured, so few of the slots either method spends are unverifiable. Burial already does reasonably here — the three strongest hot spots are among the most buried — and HotSpotter's gain comes from the weakly buried residues it promotes rather than from any failure of burial at the top of its own list.

3.8 Where the ceiling is

Residues that are near-identical across all 30 features still differ in measured $\Delta\Delta G$ by an average of **1.23 kcal/mol**, roughly 2.5 times the experimental error. Most of the remaining error is therefore not model capacity but information a single bound structure does not contain: conformational entropy, water networks, the unbound-state ensemble, allosteric context, and measurement variation between laboratories and assay conditions.

4 Discussion

4.1 What this does and does not show

It shows that a tabular model over interpretable structural features, evaluated with folds grouped by complex, substantially outperforms the burial heuristic that structural biologists use by eye, and performs comparably to published predictors that invoke more expensive physics. It does so from a single structure, in seconds, with no molecular dynamics and no free-energy calculation.

It does not show state of the art. The external comparison rests on 20 complexes and cannot separate HotSpotter from the Lise SVMs or Robetta; the one significant win is over the weakest

comparator in the set. The single modern method that would be the real test, mCSM-PPI2, cannot be compared on this data at all because of training-set overlap.

4.2 Limitations

The alanine-only label discards data. Restricting to X→Ala substitutions makes the label mean one thing but drops every complex with no alanine scan on its interface.

Coverage is partial and non-random. SKEMPI measures a median of 31% of a typical interface, and only 1 of 52 sampled complexes is scanned to ≥95%. Residues get measured because somebody thought they were interesting, so the negative class is not a random sample of non-hot residues.

Conservation depends on an external database. About 4% of chains have no ConSurf-DB entry. The model degrades gracefully — blanking conservation entirely costs only −0.0080 — but the dependency is real for a novel or designed complex.

Non-X-ray structures are not supported. The feature set includes crystallographic B-factors, which NMR and predicted structures do not have in a comparable sense; a predicted structure's B-factor column holds pLDDT, which carries the opposite meaning.

A documented failure case. On ARL15–CNNM2 (PDB 8F6D), a real target from a collaborator, burial outranks HotSpotter across all four copies in the asymmetric unit (mean rank 2.5 against 7.5). An earlier, more favourable result came from a single copy and did not survive being checked against the others. This is reported because selecting the copy that flatters the model is exactly the error the rest of the protocol is built to prevent.

4.3 Negative results

Recording what failed is as useful as recording what worked, and ten interventions were tried against the tabular baseline. Two helped: the Wide & Deep skip connection and L1 regularization. Eight did not, despite every one of them adding genuine information — feature cleanup, XGBoost stacking, ESM-2 embeddings at 320 and 650 dimensions, an ESM-derived conservation scalar at two model sizes, atom-level chemistry, PCA variants of ESM-650M, a GNN/XGBoost ensemble, and implicit-solvent electrostatics with per-atom AMBER charges and PROPKA protonation states. Separately, five feature-selection methods were tested and none beat using all features: recursive feature elimination, random-forest importance, mRMR, MRMD, and SVM-RFE.

A pattern that consistent is not eight unrelated accidents. Taken with the ceiling measurement in Section 3.8, the most likely explanation is that the features are substantially redundant restatements of burial and local packing, and that the residual error is dominated by information absent from the input rather than by anything a better model or a longer feature list would recover.

5 Conclusion

Burial is not the same as load-bearing. Ranking interface residues by buried surface area recovers 60% of complexes' top hot spot; a gradient-boosted model over 30 structural, chemical and evolutionary features recovers 78.2%, and on one case study recovers three of four hot spots where burial recovers none. The gain is robust to grouping folds by sequence identity, survives nested hyperparameter selection with a disclosed optimism of +0.008, and holds under 1 Å of simulated docking error. Against published methods on matched external rows it is competitive rather than superior, and the honest limit of that comparison is 20 complexes.

Data and code availability

Code: `https://github.com/mbrown01/hotspotter`. The training table, the fixed fold assignment and the chain sequences used for identity clustering ship with the repository, so every reported table reproduces from a clone:

- `scripts/evaluate.py` — the headline results
- `scripts/audit_splits.py` — the fold-grouping comparison
- `scripts/nested_cv.py` — the hyperparameter selection bias measurement
- `scripts/diagnose_ceiling.py` — the ceiling analysis
- `scripts/infer_xgb.py` — score a single complex

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